Breast Ultrasound Lesion Segmentation

Surabhi Fadnavis · high-school research project · model v1.0.0

What this is

A browser demo that draws a lump outline on breast ultrasound images and shows a harmless vs cancerous score. The model runs on your device. Images are not uploaded.

Key results (browser model v1.0.0)

BUSI held-out test
Outline-overlap (Dice) 0.697 · lump-only Dice 0.764 · AUC 0.931

BUS-BRA (outside)
Dice 0.714 · AUC 0.638

BrEaST (outside)
Dice 0.627 · AUC 0.721

BUS-UCLM (outside)
640 images / 38 patients · Dice 0.386 · lump-only 0.679 · AUC 0.780

Limits

Research only. Not a medical device. Scores are not a diagnosis. Outside datasets score lower. Normal (no lump) scans can still get false outlines. Do not use for care decisions.

Live demo
Scan to open the site

Credit

Project by Surabhi Fadnavis. The code, analysis, text, and video were produced with AI tools (Cursor).

Cite / DOI

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Disclaimer

This project is a student research demo. It is not for clinical use, screening, diagnosis, or any care decision.